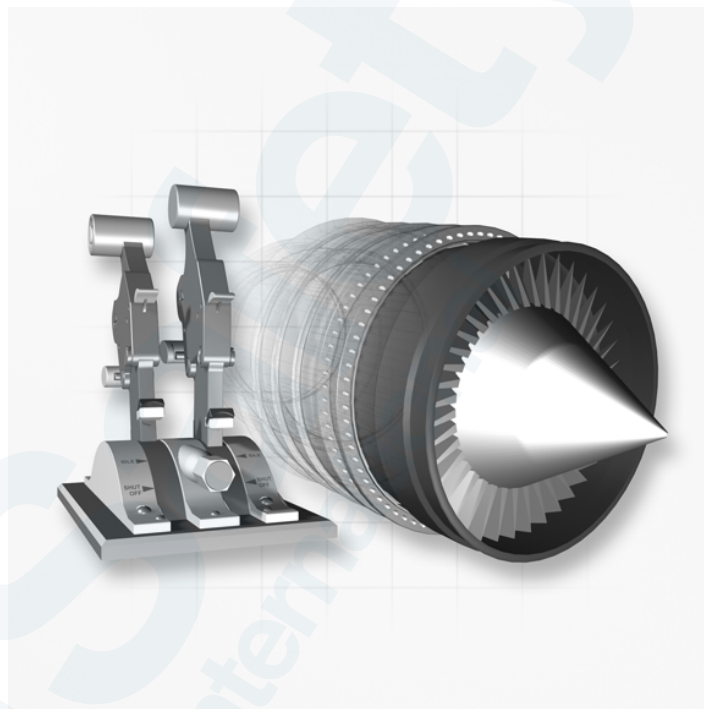


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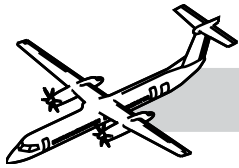
ENGINE STANDARD PRACTICES



20 STANDARD PRACTICE

70-00-00 INTRODUCTION

This chapter details the standard practices for engines, examples of which are shown in the tasks to follow.



STANDARD TORQUE FOR ENGINES

The maintenance procedure that follows is to torque nuts or bolts:

- Torque values that are given in the tasks are at room temperature
- Angles of turn are in degrees
- The torques that are given are the torques that must be applied to the part. They do not take into consideration any adapter or extension used when the part is torque
- You must lubricate the threads with engine oil, unless stated differently.

NOTE

Nuts or bolts that are silver plated do not require lubrication before assembly.

- Torque the nuts or bolts on a flange in a star shaped pattern
- Parts that are heated or cooled before assembly **MUST** be at room temperature when you apply the final torque
- Apply the torque slowly and evenly for more precision.

TEMPORARY MARKING OF PARTS

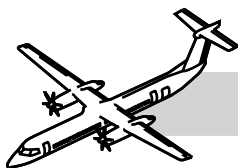
Use temporary marking methods when you need the mark to remain visible while the part is handled, stored or assembled.

CAUTION

DO NOT USE A MARKING PROCEDURE THAT PUTS LEAD, COPPER, CARBON, ZINC OR OTHER EQUIVALENT MATERIAL ON THE PART. WHEN THE PART BECOMES HOT, CARBURIZATION OR INTERGRANULAR DAMAGE CAN CAUSE IT TO HAVE A DECREASED FATIGUE STRENGTH.

The task lists a number of safe temporary marking methods as follows:

- Ink Marking
- Including ink types that are applicable for hot and cold section engine parts
- Marking materials that are applicable for cold section engine parts only
- Marking materials that are applicable for hot section engine parts only.



INSTALLATION OF THE WIRING HARNESS CONNECTORS

Do this task when it is necessary to connect an electrical connector to an engine component.

NOTE

The installation of the heat-shrink sleeve (tubing) is optional but recommended on connectors where corrosion and vibration can be a problem. The connectors are identified in the task.

There are alternate methods for protection of the connectors as follows:

- Alternate Method #1, Self-Fusing Tape
- Alternate Method #2, High Temperature Heat Shrink Tape
- Clean the connector with a diluted mixture of contact enhancer
- Connect the connector to the mating receptacle and tighten the locknut lightly by hand
- With your hand, apply a small counterclockwise load on the backshell.
- Torque the connector locknut clockwise until you cannot see the red color band in the mating receptacle (With the counter-clockwise load on the backshell, torque the connector locknut clockwise to 100 lbf-in (11.3 N-m).

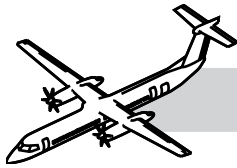
NOTE

A wrench should always be used to tighten connectors. Not doing so will lead to connectors becoming loose and electrical circuit malfunction.

RECOMMENDED ENGINE OPERATIONAL PRACTICES FOR HARSH CLIMATES

For engines exposed to high ambient temperature and to reduce thermal stress on hot section components, P&WC recommends the following:

- Start using battery cart/APU power at all locations
- To cool the engine, motor the engine for 10-15 seconds prior to selecting CLA to the START/FEATHER position, if the engine temperature before starting is >200 degrees C
- Cool down engines before shutdown
- Use RDC TOP where possible
- Transition from NTOP/MTOP to MCL as soon as possible
- Operate at reduced Climb and Cruise power where possible
- Bleeds selected off for Take-Off
- Avoid use of Reverse Power on landing (except in emergency)
- Monitor bleed valve operation based on ECTM results
- When bleed leaks are suspected perform Power Assurance Checks with bleed ports capped and uncapped
- Check airframe bleed system for leaks on a regular basis.



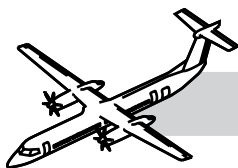
For engines subject to sand erosion and FOD, P&WC recommends the following:

NOTES

- Restrict CLA movement when the aircraft is stationary
- Taxi using both engines with equal PLA movement
- Use intake blanking covers when the aircraft is parked overnight
- Avoid using reverse on landing (except in emergency)
- Avoid running into sand wakes during taxi.

For engines exposed to severe marine environment and/or volcanic fumes P&WC recommend the following:

- Perform compressor/turbine wash every 50FH. Adjust washing interval based on level of exposure to the salt laden atmosphere
- Perform an initial borescope inspection on the HP, LP and PT sections at 2000FH TSN/TSO for evidence of sulphidation attack on the turbine blades. Adjust inspection interval as necessary
- Perform external engine washes every 1200FH to reduce salt deposits on the engine
- Perform an initial visual inspection of all magnesium parts at 1200 hours TSN/TSO
 - A. Inspect external surfaces and inlet gas path for corrosion or damage
 - B. Repair magnesium surfaces i.a.w. the AMM . After treatment apply 2 or 3 coats of paint
 - C. Pay particular attention to bolted flanges or other areas where the painted surface may be compromised.



70-00-00 SPECIAL TOOLS & TEST EQUIPMENT

- PWC57689 Air Deflector Shield
- PWC58104 Wrench, mini-strap
- Glenair TG70 Wrench, mini-strap
- Glenair TG69 Pliers, soft-jawed
- GSB2000020 or equivalent Variable temperature, electronically controlled heat gun (120 Vac - 60 Hz)
- GSB2000022 or equivalent Variable temperature, electronically
- Controlled heat gun (230 Vac - 50 Hz)
- PNo. 07051 (39 mm) (Steinel) or equivalent Heat gun reflector nozzle
- GSB2400016 Tool kit, electrical and avionics